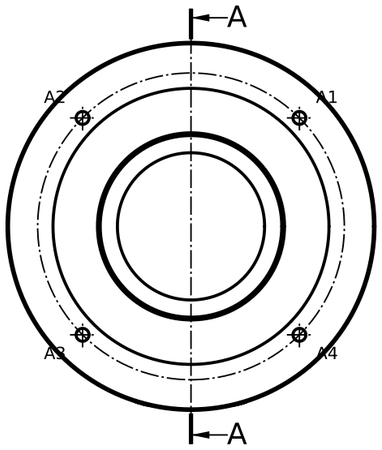
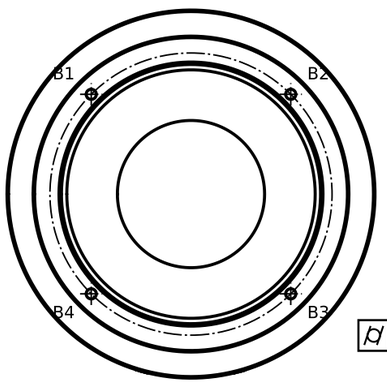
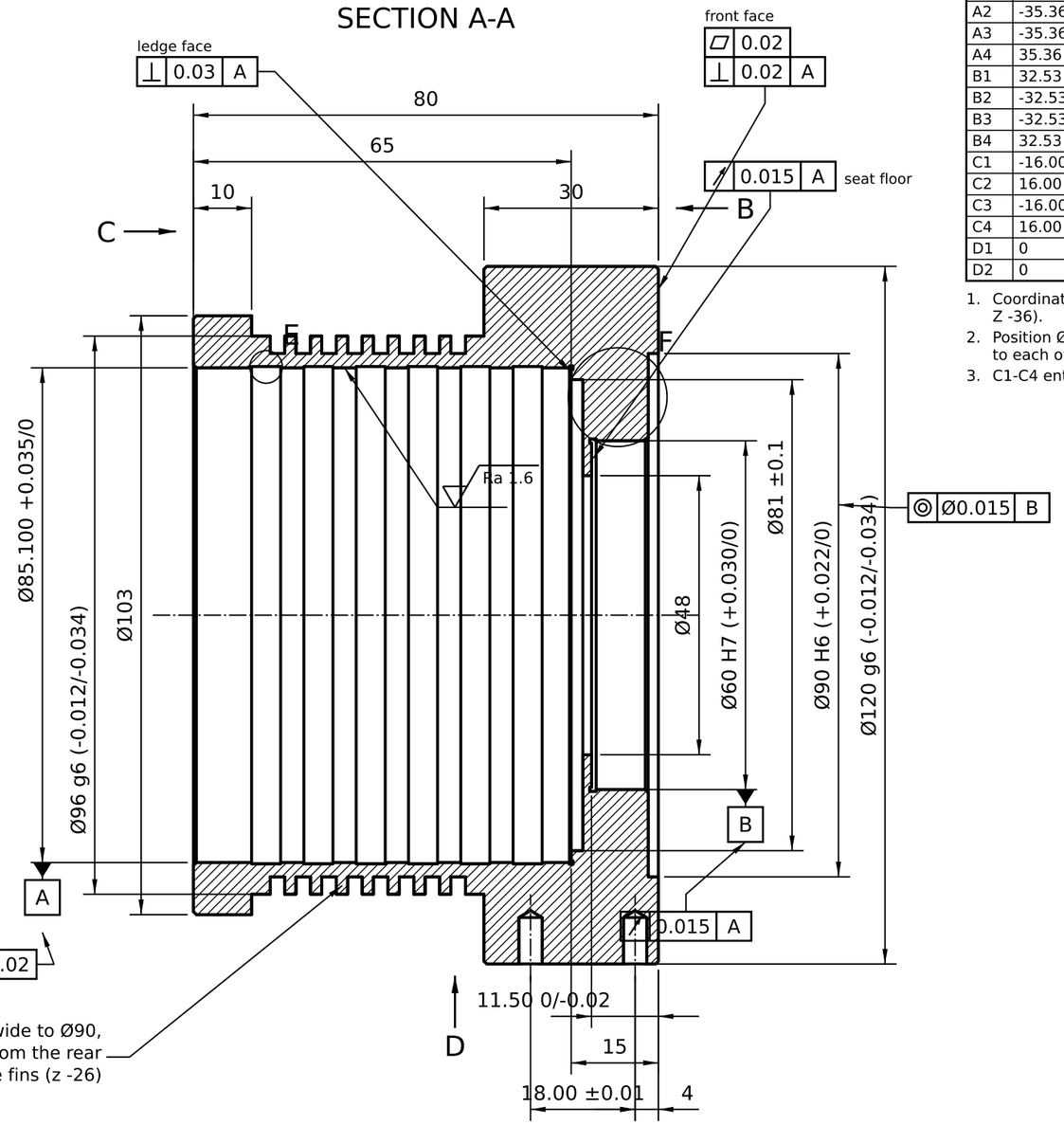
VIEW B (1:2)



VIEW C (1:2)



SECTION A-A



TAG	X	Y	Z	HOLE	THREAD / DRILL	FROM
A1	35.36	35.36	44	M5-6H	15 / Ø4.2×18	front face
A2	-35.36	35.36	44	M5-6H	15 / Ø4.2×18	front face
A3	-35.36	-35.36	44	M5-6H	15 / Ø4.2×18	front face
A4	35.36	-35.36	44	M5-6H	15 / Ø4.2×18	front face
B1	32.53	32.53	-36	M4-6H	9 / Ø3.3×10.5	rear face
B2	-32.53	32.53	-36	M4-6H	9 / Ø3.3×10.5	rear face
B3	-32.53	-32.53	-36	M4-6H	9 / Ø3.3×10.5	rear face
B4	32.53	-32.53	-36	M4-6H	9 / Ø3.3×10.5	rear face
C1	-16.00	-57.83	28.00	M5-6H	12 / Ø4.2 to Y-43.5	Ø120 (spot)
C2	16.00	-57.83	28.00	M5-6H	12 / Ø4.2 to Y-43.5	Ø120 (spot)
C3	-16.00	-57.83	37.00	M5-6H	12 / Ø4.2 to Y-43.5	Ø120 (spot)
C4	16.00	-57.83	37.00	M5-6H	12 / Ø4.2 to Y-43.5	Ø120 (spot)
D1	0	-60	22.00	Ø4 H7	6 min / Ø4×8	Ø120
D2	0	-60	40.00	Ø4 H7	6 min / Ø4×8	Ø120

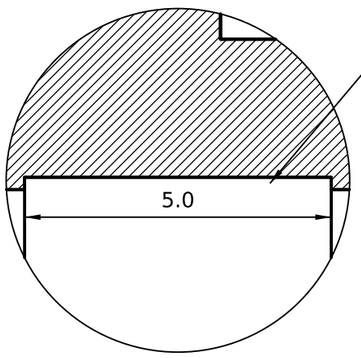
1. Coordinates in the STEP frame: Z axis = datum A, Z0 as the model (front face Z 44, rear face Z -36).
2. Position Ø0.2 to A for A1-A4, B1-B4, C1-C4. D1-D2: Ø4 H7 (+0.012/0), position Ø0.02 to A and to each other (±0.01), dowels ISO 8734 4 m6 slip here.
3. C1-C4 enter the Ø120 at 15.5°: spot face with a Ø5 flat end mill before drilling.

NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/housing.step (STEVE 02e31a10 rev 49).
3. Edges ISO 13715 -0.3 (break 0.2-0.4 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Threads ISO metric coarse, 6H internal / 6g external; thread depths are full-thread depths.
6. Material EN AW-6061-T6 (T6511 extruded bar acceptable), with certificate.
7. Black anodise MIL-A-8625 Type II Class 2, 8-12 µm. MASK (chromate MIL-DTL-5541 Type II Class 3 instead): Ø85.100 bore and grooves, Ø60 seat and floor, Ø90 counterbore, Ø120 and Ø96, tapped holes (or chase after anodising), dowel holes. Fits apply after finishing.
8. Turn Ø60, its floor and Ø90 in one setup. Ø60 seat and its floor Ra 0.8. Stator bore Ra 1.6 (not polished: bonded joint).
9. Chamfers 0.5×45°: stator-bore entry (rear), Ø60 seat entry, Ø120 front edge, Ø120/Ø96 step, Ø103 rear edge.
10. Inspect: CMM bores and runouts to A, front-face flatness, hole positions; groove depth by gauge or replica; thread gauges after anodise; dowels (AM-05) must drop in and lift off freely.

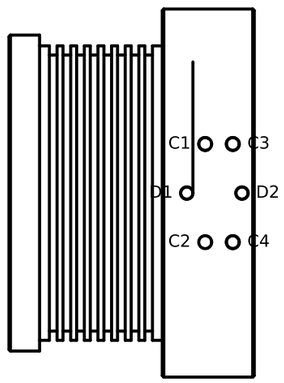
Ra 3.2 (√ Ra 1.6, Ra 0.8)

DETAIL E (10:1)

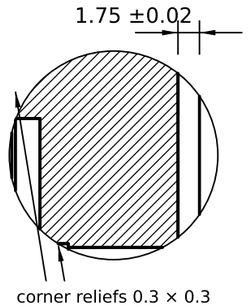


6× adhesive grooves 5.0 wide,
0.20 ±0.05 deep (Ø85.50 ±0.10),
pitch 9.00, from z -26 to 24
(first groove 10.0 from rear face)

VIEW D (1:2)



DETAIL F (2:1)



REV	DESCRIPTION	DATE	BY
B	Ø90 counterbore H6, 1.75 deep (was H7, 1.60)	2026-10-17	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash		SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 49 STEP: cnc/step/housing.step	
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081) Edges ISO 13715 -0.3 (0.2-0.4)	PROJECTION: FIRST ANGLE ISO 128 / ISO 5456-2			
FINISH / HEAT TREATMENT: Black anodise Type II, fits masked (chromate)				UNITS: mm
MATERIAL: EN AW-6061-T6				SCALE: 1:1 (views as noted)
TITLE: **Motor housing**				QTY PER ACTUATOR: **1**
DRAWING / PART NUMBER: **AM-01-housing**	REV: **B**	SHEET: 1/1	SIZE: A3	DATE OF ISSUE: 2026-10-17